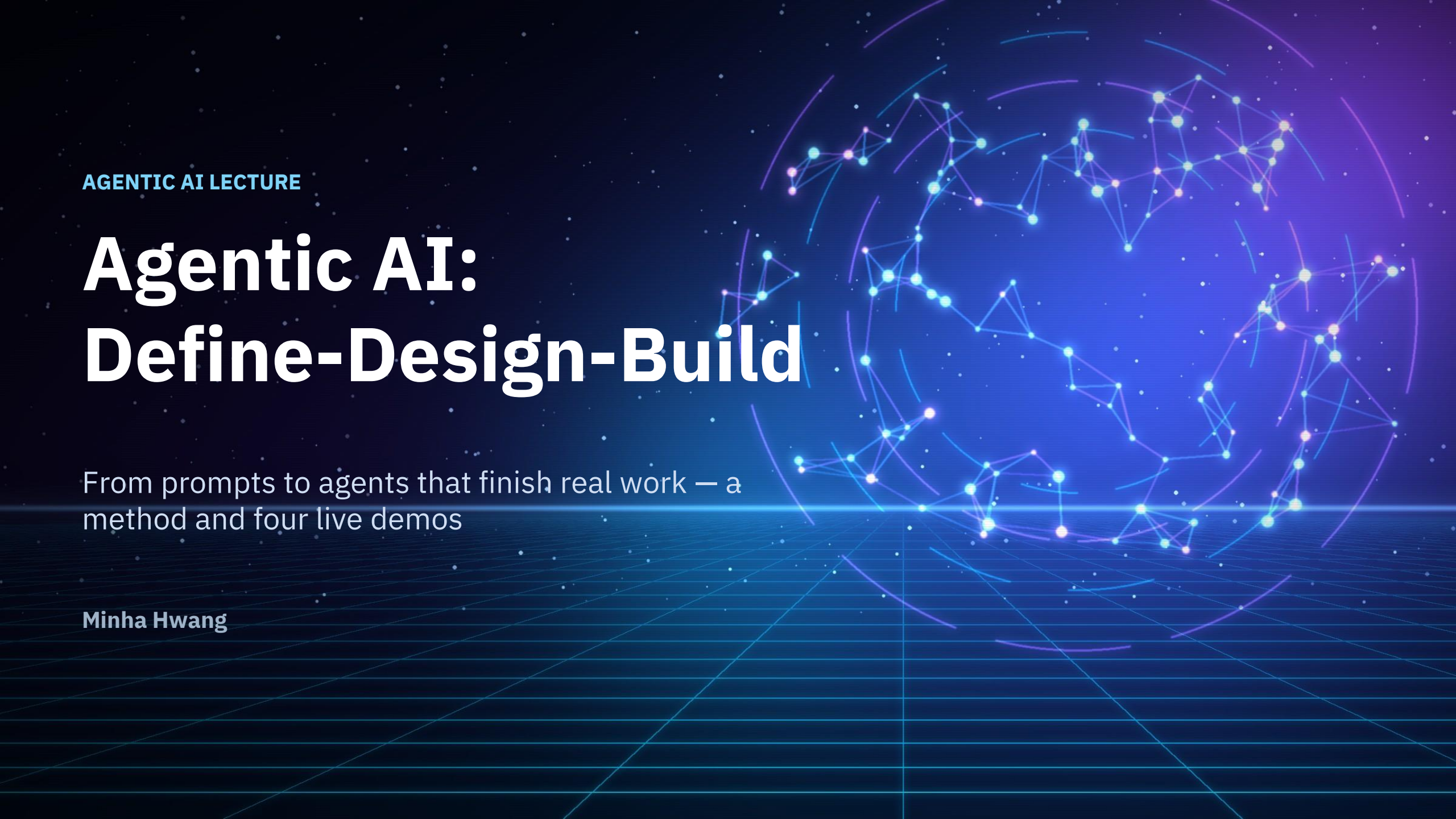
AGENTIC AI LECTURE

Agentic AI: Define-Design-Build

From prompts to agents that finish real work — a method and four live demos

Minha Hwang



AGENDA



01 **Intro to Agentic AI**
From chat, to coding agents, to knowledge-work agents

02 **The method: Define → Design → Build**
Seven decisions behind every useful agent

03 **Live demos**
Competitive intel · Pitch-book panel · DCF valuation · Annual-report dashboard

04 **Vibe coding → Agentic engineering**
Building apps from a single prompt

05 **Final thoughts**
Learning by doing, and the human skills that matter more

INTRO TO AGENTIC AI

Generative AI and Agentic AI Evolution

The scope of delegation expands from answers to software to knowledge work

2022–2024

Chat + search

ChatGPT moment

Ask questions and find information

ChatGPT: 30 Nov 2022
ChatGPT Search: 31 Oct 2024

2025

Coding agents

Claude Code + Opus 4.5

Inspect a repo, edit code and run tests

Claude Code preview: 24 Feb 2025
Opus 4.5 model: 24 Nov 2025

EARLY 2026

Knowledge work agents



OpenClaw



Claude Cowork

Delegate work across files and connected apps

Cowork preview: 12 Jan 2026
OpenClaw rebrand: late Jan 2026

The unit of interaction expands from an answer to a completed task

The Proliferation of AI Agents

Personal agents

Personal work, research and creation



OpenClaw



Claude Cowork



Genspark



Hermes Agent



Muse

Meta



Dots

OpenAI



Manus



Perplexity Computer

Enterprise agents

Company context and shared workflows



Microsoft Copilot

Cowork / Autopilot*



ChatGPT Work

OpenAI



Claude Cowork

Enterprise



Gemini Enterprise

Google



Agentforce

Salesforce



Glean Agents

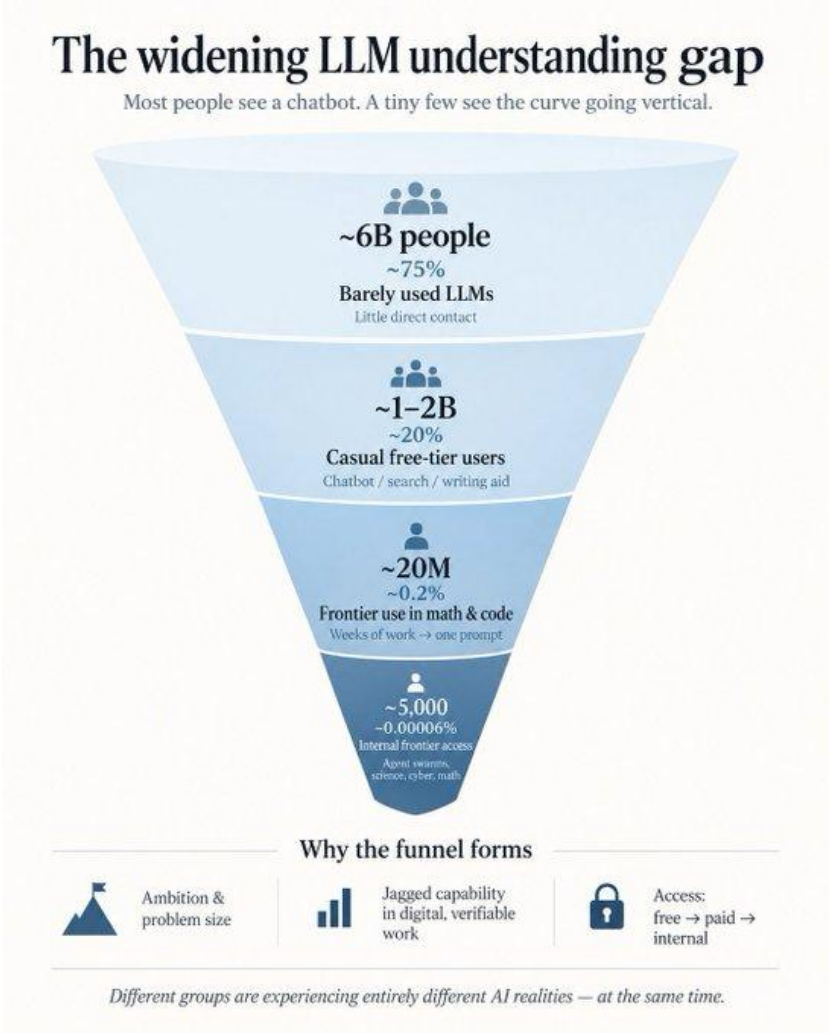


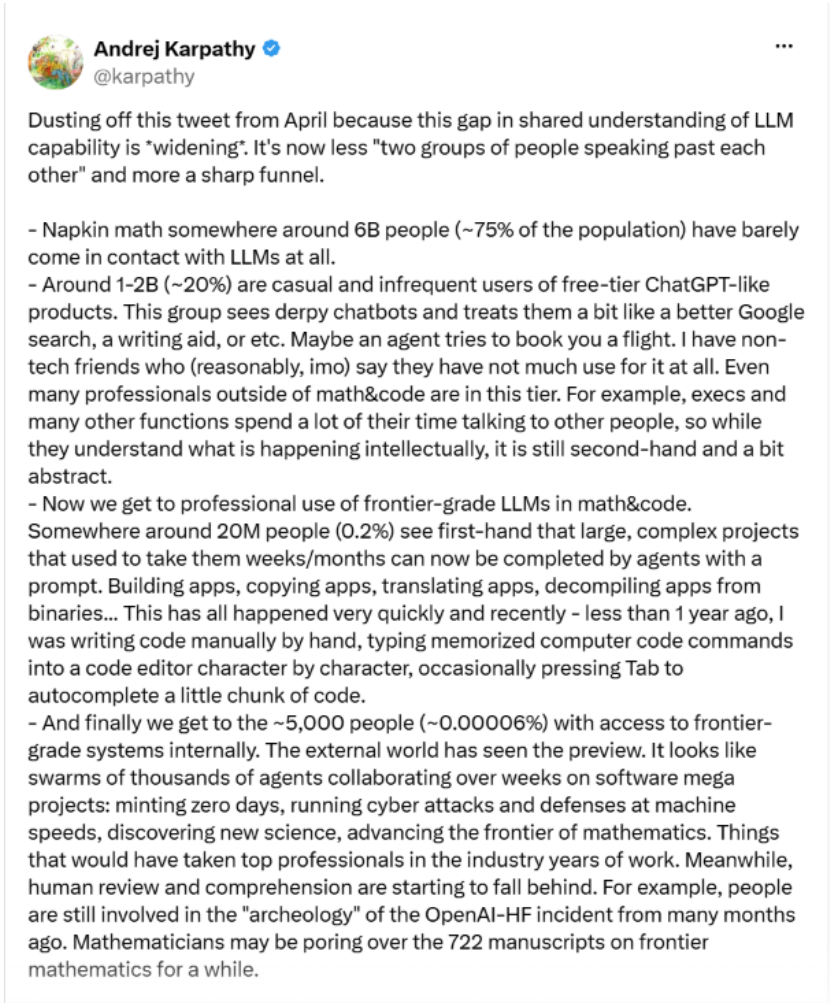
ServiceNow Otto

NVIDIA **NemoClaw** Agent blueprints and runtime controls

Examples span personal and enterprise use. *Autopilot: private preview.

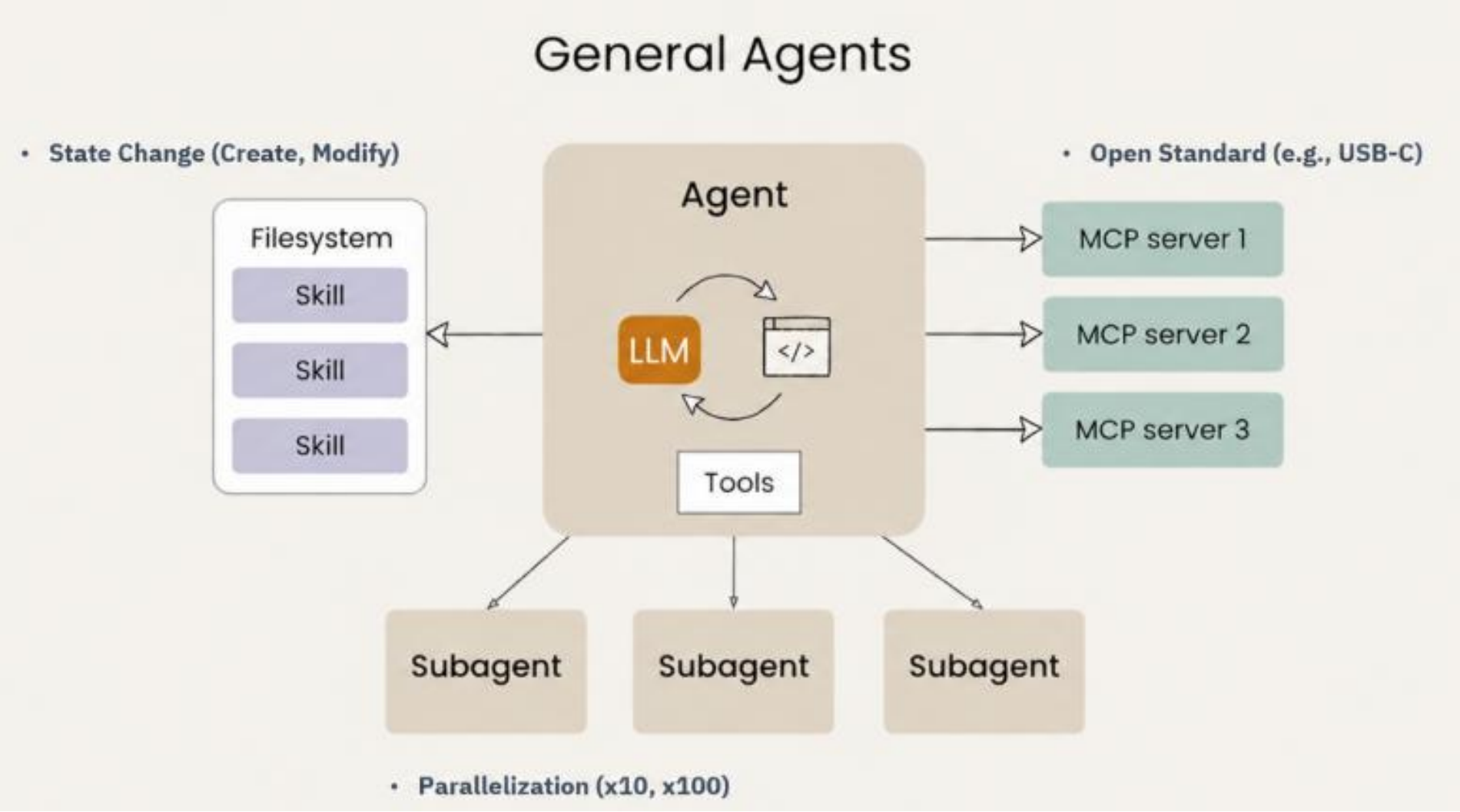
The Widening Agentic AI Adoption Gaps





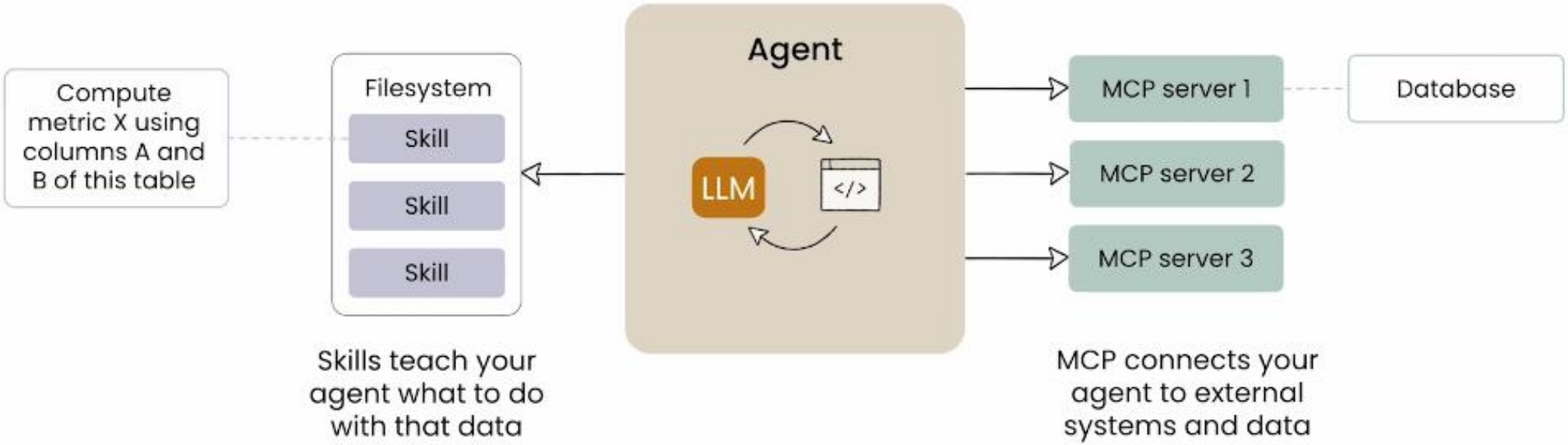
Andrej Karpathy on X: "Dusting off this tweet from April because this gap in shared understanding of LLM capability is *widening*. It's now less "two groups of people speaking past each other" and more a sharp funnel. - Napkin math somewhere around 6B people (~75% of the population) have barely co… / X

Agents, Tools, Skills, and MCP (Model Context Protocol)



Source: https://learn.deeplearning.ai/courses/agent-skills-with-anthropic

Skills vs MCP

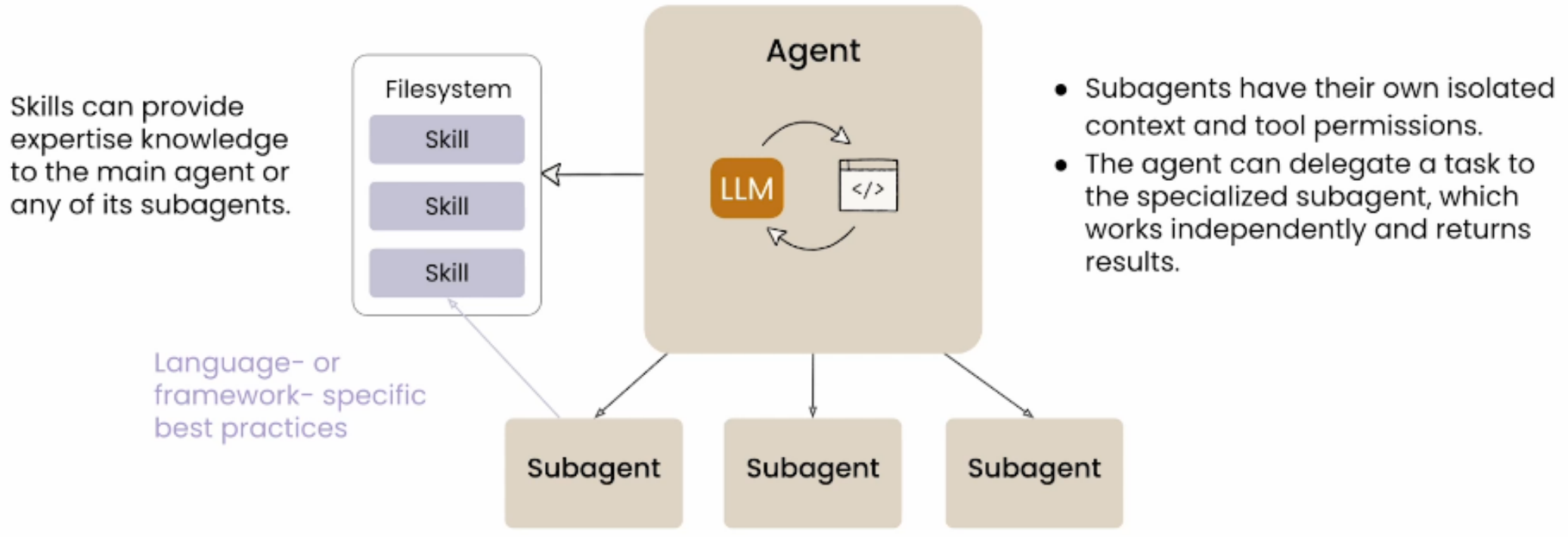


- Workflow, which are repeatable
- **Set of instructions** to put tools together
- Bringing in underlying tools and resources when needed
- Lazy memory (invoked)

- **Open standard** (like USB-C)
- Helps LLM to understand how to use **external APIs** (e.g., web search)
- Bring in external data

Source: https://learn.deeplearning.ai/courses/agent-skills-with-anthropic

Subagents



Story reviewer Number reviewer Devil's advocate

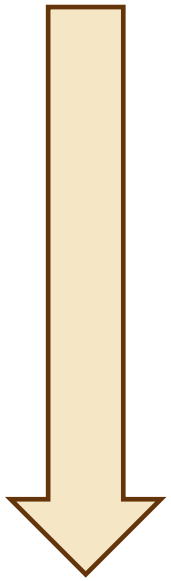
Subagents report results back to the main agent. Sub agents are specialists, which can be run in parallel

Source: https://learn.deeplearning.ai/courses/agent-skills-with-anthropic
Orchestrate teams of Claude Code sessions - Claude Code Docs

Agentic AI: Essay Writing Example

Non-agentic workflow (zero-shot)

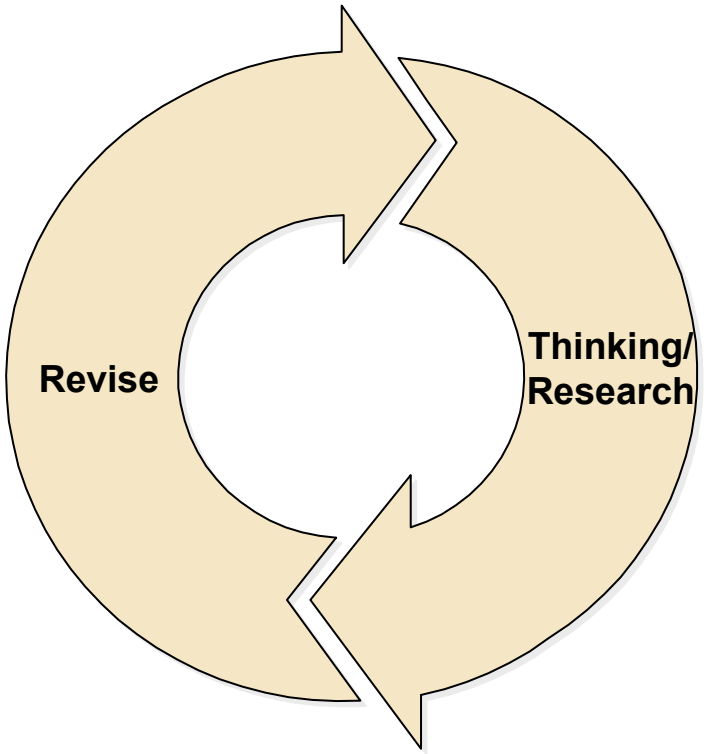
Start



End

Please type out an essay on topic X from start to finish in one go, without using backspace.

Agentic workflow



Write an essay **outline** on topic X

Do you need any **web search (tool)**?

Write a **first draft**

Consider what parts need **revision or more research**

Revise your draft

…

- **Multiple steps** (**parallelization** possible)
- **Reflection loops**
- (Multiple) **tool use**
- Conversational UI

Source: Agentic AI - DeepLearning.AI

Agentic AI Workflows

An agentic AI workflow is a process where an LLM-based app executes **multiple steps** to **complete a task**

Essay-writing example:

Write an essay outline on topic X

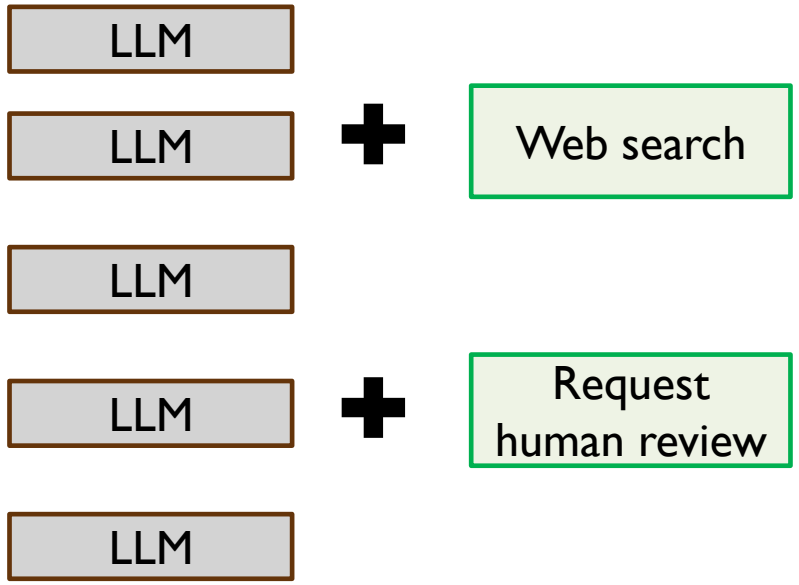
Do you need any web search (tool)?

Write a first draft

Consider what parts need revision or more research

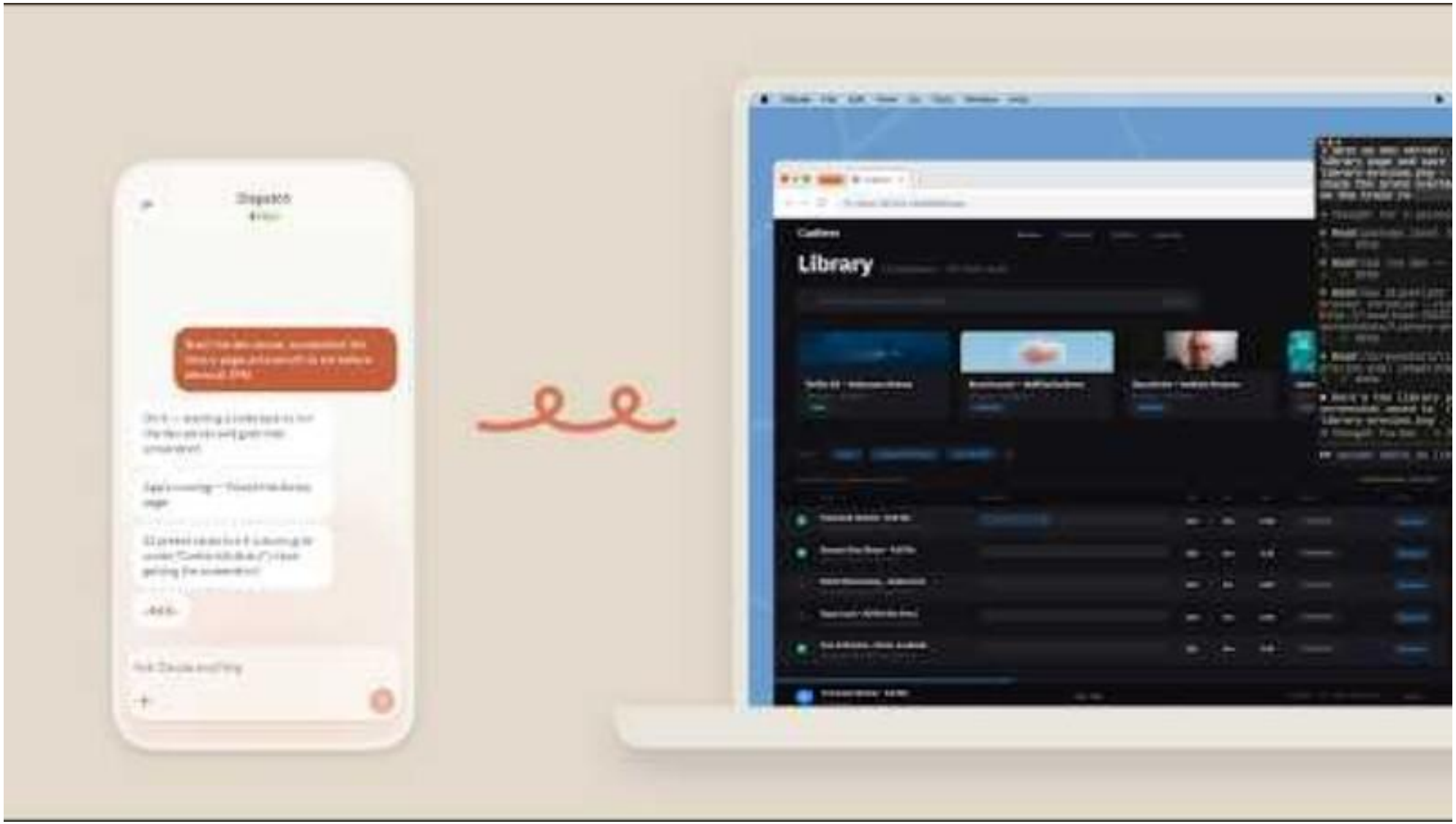
Revise your draft

…



Source: Agentic AI - DeepLearning.AI

AGENTIC AI: CLAUDE COWORK (1/3)



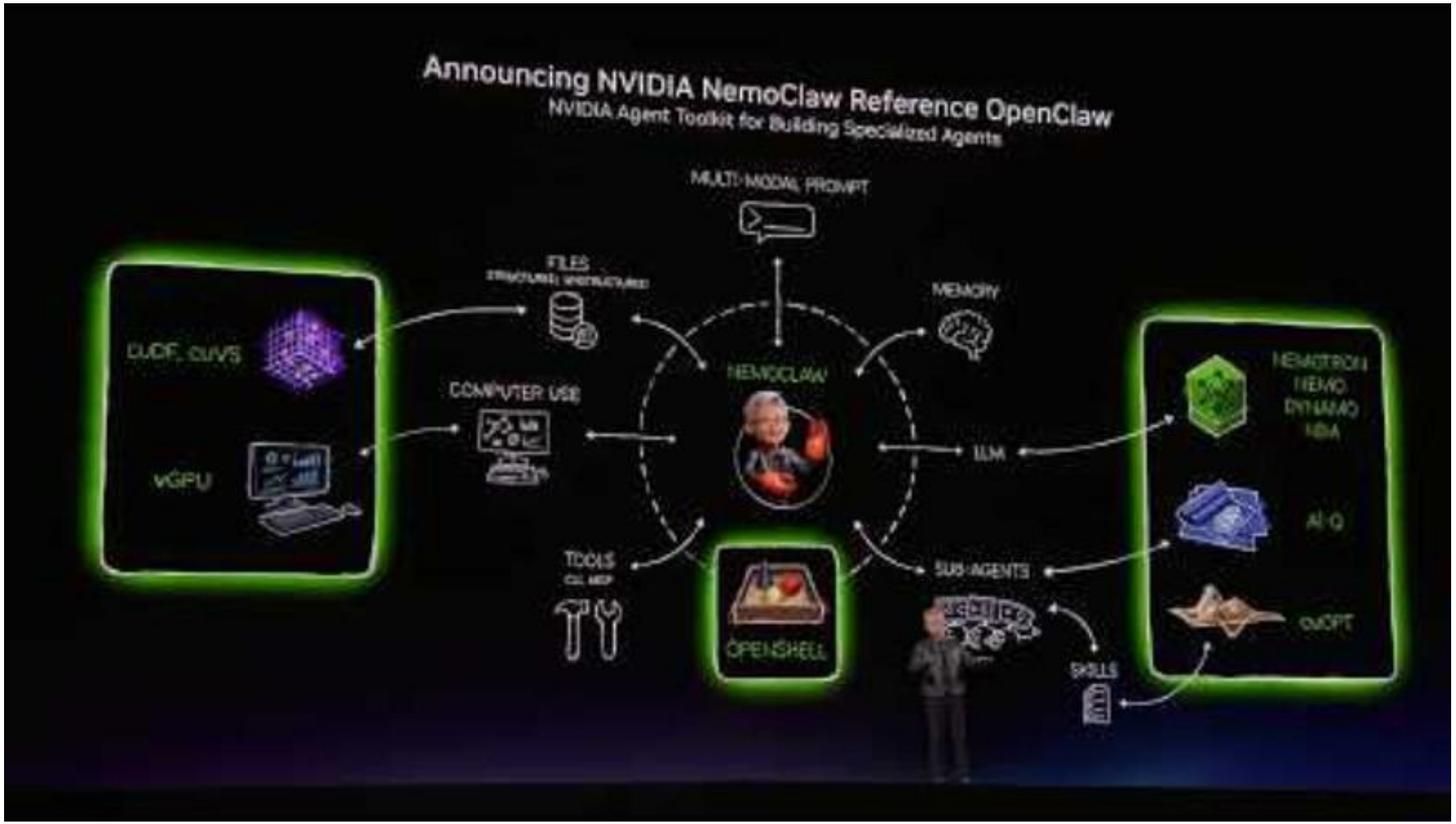
Source: [Claude Cowork | Anthropic's agentic AI for knowledge work \ Anthropic]

AGENTIC AI: COPILOT COWORK (2/3)



Source: [Copilot Cowork: A new way of getting work done | Microsoft 365 Blog]

AGENTIC AI: OPENCLAW (3/3)



Source: [NVIDIA's Jenson Hwang launches NemoClaw to the OpenClaw community]

THE METHOD

Define → Design → Build: seven decisions

DEFINE (PROBLEM)

1 · Task

Whose decision gets better? What is painful today?

2 · Success

What does good look like, and when must it land?

DESIGN

3 · Sources

Where do the relevant facts live?

4 · Skills

Which built-in skills? What know-how must be written down?

5 · Design Pattern

Single pass, routine, pipeline, fan-out or panel?

6 · Outputs & gates

What format, for whom, and what needs my OK?

BUILD

7 · Build loop

Prompt → run → refine → save as skill → automate → share

5 · PATTERN

Five ways to split the work

Design Pattern	Shape	Use it when	Today
Single pass	one prompt → one output	The job fits in one read and one write	Demo 3 (DCF); where every agent starts
Routine	trigger → run → deliver	It must happen on a cadence or when something arrives	Demo 1, automated monthly
Pipeline	A → file → B → file → C	The source is too big for one pass, or steps depend on each other	Demo 4
Fan-out	same brief × N → synthesis	Independent pieces need identical treatment	Demo 1
Panel	lenses → challenge → verdict	You want disagreement surfaced, not consensus	Demo 2

7 · BUILD

Every prompt has six parts

Role	Who it is working as
Goal	The outcome, in one sentence
Sources	Exactly where to look, and the time window
Steps	The order of work; what to judge and how
Output	Format, name, destination
Guardrails	What it must not do; what to flag

ROLE: You are an equity research analyst.

GOAL: Value Alphabet (GOOGL) today with a model I can audit.

SOURCES: Alphabet 10-K and 2026 earnings releases (SEC); latest capex guidance; peer multiples (MSFT, META, AMZN).

STEPS: Historicals → segment projections 2026–30 → DCF + sensitivity → multiples → summary.

OUTPUT: output/GOOGL_DCF_Valuation.xlsx — blue inputs, formulas only.

GUARDRAILS: Cite a source for every input. Show a range, not a single target price.

7 · BUILD

The build ladder: from one prompt to a routine

1 · Prompt once
Run it by hand and read the result

→

2 · Refine
Correct it in the same task until it's right

→

3 · Save as skill
Customize › Skills, or 'turn this into a skill'

→

4 · Automate
A scheduled prompt or an event trigger

→

5 · Share
Share the skill with named colleagues

Rule of three: run it by hand three times before you automate it.

LIVE DEMO


AGENTIC COWORK HUB
COLLABORATE SMARTER
AGENTIC COWORK SOFTWARE | INTEGRATED WORKSPACE
CLAUDE COWORK
Microsoft Copilot Cowork
PERPLEXITY COMPUTER
GENSPARK
OPENCLAW
MANUS
LIVE AGENT ACTIVITY
PROJECTS IN PROGRESS
Sprint Planning
Data Analysis
Content Strategy
COLLABORATION HUB
Chat
PERFORMANCE METRICS

DEMO 1 · FAN-OUT · BEATS 1–2

Competitive intelligence: five robotics startups

DEFINE · 1 · TASK

Before the Q4 investment committee we need a like-for-like view of five leading embodied-AI startups, not five differently shaped memos.

DEFINE · 2 · SUCCESS

Five profiles with identical headings, every fact dated and sourced, plus a differentiator summary. Refreshed monthly and after any funding event.

DESIGN · 3 · SOURCES

Web: company blogs and press releases · SEC filings (Agility S-4) · funding announcements · tech and business press · model papers and release notes

DESIGN · 4 · SKILLS

Built-in: Deep Research, Excel, Word, PowerPoint · Custom: startup-profile (headings, source rules, “reported” labels)

DESIGN · 5 · PATTERN

Fan-out: five parallel sub-agents on the same brief, then one synthesis.

DESIGN · 6 · OUTPUTS & GATES

One profile per company · research/summary.md · competitor-matrix.xlsx · three slides for the IC

DEMO 1 · WORKFLOW

Same brief five times, then synthesise

One brief
Five headings, 12 months, source rules

→

5 SUB-AGENTS

- **Figure AI**
- **Physical Intelligence**
- **Skild AI**
- **Agility Robotics**
- **1X Technologies**

→

Five profiles
One Markdown file per company

→

Synthesise
Summary + Excel matrix

→

IC pack
Three slides; you review

IDENTICAL BRIEFS
Identical headings make the profiles comparable.

SPEED
Five pieces of independent research run side by side.

INSIGHT IS IN THE SYNTHESIS
Differentiators and implications only appear when compared.

DEMO 1 · BEAT 3 · BUILD PROMPT

Five researchers, one brief

Research these 5 robotics startups in parallel, using a separate sub-agent for each and the same brief.
Sub-agent 1: Figure AI · Sub-agent 2: Physical Intelligence · Sub-agent 3: Skild AI
Sub-agent 4: Agility Robotics · Sub-agent 5: 1X Technologies

BRIEF: Last 12 months, under these exact headings:
1 Funding & valuation: rounds, lead investors, post-money valuation, total raised, IPO/SPAC moves
2 Technology: robot hardware and/or foundation model (VLA, world model), key releases, open source
3 Commercial traction: named customers, deployments, revenue or ARR, orders, pricing
4 Manufacturing & scale: factories, units built or delivered, capacity targets
5 Team, partners & risks: leadership changes, partners, lawsuits, safety or teleoperation concerns

RULES: Cite and date every fact. Prefer company announcements and filings to media. Label unconfirmed items "reported". Flag anything older than 12 months.

SAVE: Each profile to research/{company}.md.

SYNTHESIS: Once all 5 finish, write research/summary.md: a side-by-side table, 5 key differentiators, who is gaining or losing ground, and 3 implications for our investment committee. Also build research/competitor-matrix.xlsx.

DEMO 1 · BEAT 5 · HARDEN

Turn the research into a pack, then a routine

NEXT PROMPT · THE IC PACK

“Turn research/summary.md into 3 slides for the investment committee: the landscape, the differentiators, and the implications for our fund. Use our PowerPoint template.”

AUTOMATE · FIRST MONDAY OF EACH MONTH

“On the first Monday of each month at 7am, rerun the startup-profile brief and put what changed since the last summary at the top: new rounds, customers, launches, leadership moves.”

SAVE AS SKILL

startup-profile: the five headings, the 12-month window, the source rules. Reuse it for AV, drone or AI-chip startups.

BEFORE IT REACHES THE IC

Spot-check three facts per profile. Treat “reported” valuations as unconfirmed until a round closes.

DEMO 1 · BEAT 4 · RESULTS

Five profiles, one side-by-side view

Company / layer	Financing reference	Commercial evidence
Figure AI · humanoid	$39B post-money, Sep 2025 Historical benchmark	BMW confirms pilot contribution; revenue not disclosed.
Physical Intelligence · model	$5.6B investor baseline, Nov 2025 Valuation basis unspecified	API beta in October 2026; revenue not disclosed.
Skild AI · model + operations	>$14B valuation, Jan 2026	>$100M ARR; 60+ customers. Company-reported, unaudited.
Agility Robotics · logistics	$2.5B proposed SPAC Vote scheduled 2 Nov 2026	$1.782M FY2025 net sales; conditional, multi-year order value.
1X Technologies · home	~$6B reported acquisition talks Unconfirmed	Preorders; external deliveries not independently verified.

WHAT THE COMPARISON SHOWS

Use comparable evidence for adoption and economics. Historical rounds, proposed transactions and reported talks have different bases. Capacity, production, orders and paid deliveries are separate measures.

Sources: company releases, counterparties, SEC filings and attributed reports in the five profiles. Cutoff: 17 Oct 2026. Teaching material.

DEMO 2 · PANEL · BEATS 1–2

Multi-agent pitch-book review

DEFINE · 1 · TASK

A partner has 20 minutes before a founder meeting and needs a sharp, balanced read of a Series A deck — not one reviewer's opinion.

DEFINE · 2 · SUCCESS

Every numeric claim checked against the data room; disagreements surfaced, not averaged away; scorecard, top 5 fixes and a go/no-go on one page.

DESIGN · 3 · SOURCES

pitch-deck.pdf (11 slides) · data-room/kpi-monthly.csv (13 months of KPIs) · our Series A rubric. All data fictional.

DESIGN · 4 · SKILLS

Built-in: PDF, Excel · Custom: pitch-review (rubric, benchmarks) + three reviewer agents

DESIGN · 5 · PATTERN

Panel: three reviewers in parallel who message each other to challenge findings; a lead writes the verdict.

DESIGN · 6 · OUTPUTS & GATES

output/pitch-review.md + one file per reviewer. Nothing goes to the founders; the partner decides.

DEMO 2 · BEAT 3 · BUILD PROMPT

Agent team: three reviewers who argue

Create an agent team to review GraspLoop's Series A pitch deck (pitch-deck.pdf) against its data room (data-room/kpi-monthly.csv). Spawn three reviewers:

- **Narrative reviewer**: assess story flow, clarity, investor hook
- **Numbers reviewer**: recompute every claim from the data room; flag weak or missing data
- **Devil's advocate**: argue why an investor would pass (be harsh)

Have them message each other with conflicts (e.g., if the narrative praises a claim the numbers reviewer finds unsupported).

OUTPUT: Write a consolidated critique to output/pitch-review.md: a 1–5 scorecard, a claim-check table, a conflicts log, top 5 fixes ranked by investor impact, and a go/no-go for the partner meeting.

Plugin: agents/narrative-reviewer.md · agents/numbers-reviewer.md · agents/devils-advocate.md · skills/pitch-review

DEMO 2 · BEAT 4 · RESULTS

The panel separates growth from unsupported claims

Deck claim	Data-room finding	Interpretation
$2.4M ARR	$1.932M paying; $2.436M includes unpaid pilots	Separate paying and pilot amounts
8× YoY growth / 14 customers	6.57× paying growth; 11 paying + 3 unpaid pilots	Qualify the scope
99.7% success / 160% NRR	97.2% fleet; 99.7% best site. NRR 157.14%, n=3	State population and cohort
68% GM / 4-month CAC	41% September GM; Q3 payback proxy 26.27 months	Cohort CAC remains unverified
24-month runway	8.46 months pre-raise; 33.67 with immediate raise, flat burn	Forward cash plan is unverified

DECISION · CONDITIONAL FOUNDER MEETING

Strong paying growth merits diligence. Reconcile metric definitions, acquisition economics and the funded deployment plan before investment approval. Flat-burn ratios do not establish a forward runway.

Sources: fictional GraspLoop deck pp 6-7, 10-11; September 2025-September 2026 KPI data. All company data are invented for teaching.

DEMO 3 · SINGLE PASS + SKILL · BEATS 1–2

DCF valuation: Alphabet (Google)

DEFINE · 1 · TASK

Before our tech-sector review, I need an independent view of what Alphabet is worth now that AI capex has more than doubled.

DEFINE · 2 · SUCCESS

A working Excel model: every input blue and sourced, every output a formula, a sensitivity grid and peer multiples — in under 20 minutes.

DESIGN · 3 · SOURCES

Alphabet 10-K and 2026 earnings releases (SEC) · Q2 2026 call: capex guidance, Cloud backlog · market data and peer multiples

DESIGN · 4 · SKILLS

Built-in: Excel, web search · Custom: dcf-model (tab structure, colour code, terminal capex, checks)

DESIGN · 5 · PATTERN

Single pass: one prompt, one workbook. The skill carries the know-how.

DESIGN · 6 · OUTPUTS & GATES

output/GOOGL_DCF_Valuation.xlsx + valuation notes. I review the drivers before anyone sees a number.

DEMO 3 · BEAT 3 · BUILD PROMPT

One prompt, one auditable model

Create a comprehensive financial model for Alphabet (GOOGL) with:

- Historical financials FY2023–FY2025 and H1 2026 from SEC filings
- 5-year projections (2026–2030) by segment: Search, YouTube, Network, Subscriptions & devices, Google Cloud, Other Bets
- DCF valuation (WACC, terminal growth) with sensitivity analysis
- Multiple valuation methods (P/E, EV/Sales, EV/EBITDA vs. MSFT, META, AMZN)
- All formulas in Excel format; include assumption inputs I can modify (blue font)
- Use the latest 2026 capex guidance; cite a source for every input

OUTPUT: Write an Excel file to output/GOOGL_DCF_Valuation.xlsx

Plugin: skills/dcf-model — tabs: README · Assumptions · Summary · Historicals · Projections · DCF · Sensitivity · Multiples

DEMO 3 · BEAT 4 · RESULTS

Alphabet: value depends on the method and assumptions

VALUE PER COMMON SHARE



DCF sensitivity: $158.92–$312.31

Methods are separate comparisons; they are not blended into a target.

WHAT DRIVES THE BASE CASE

Revenue $402.8B (2025) to $880.1B (2030E); Cloud 36.1% of revenue.

Capex $200B in 2026E and $271.6B in 2027E. FCF turns positive in 2028.

Discounting WACC 9.75%; terminal growth 3.0%. Terminal value is 94.0% of EV.

EBITD EBIT + PP&E depreciation. Intangible amortization stays expensed.

TRY THE ASSUMPTION

In Excel, change 2027 capex from 46% to 36% of revenue: DCF becomes $217.17. Other drivers stay fixed.

Sources: dated workbook; SEC/issuer financials through June 2026; 16 Oct 2026 quotes and adjusted EPS consensus. Teaching model, not investment advice.

DEMO 4 · PIPELINE · BEATS 1–2

Annual report to personal dashboard: NVIDIA FY2026

DEFINE · 1 · TASK

NVIDIA's FY2026 10-K is ~100 dense pages, and every reader — board member, CFO, investor — needs a different handful of numbers.

DEFINE · 2 · SUCCESS

Within 30 minutes: a verified KPI table, trends and anomalies, a one-page summary, and a dashboard for my role. Every figure has a page number.

DESIGN · 3 · SOURCES

10K-NVDA.pdf (FY2026, filed 25 Feb 2026) · Q4 FY26 earnings release for guidance · last year's 10-K (optional)

DESIGN · 4 · SKILLS

Built-in: PDF, Excel, Word, App · Custom: semis-annual-report-kpis (platforms, margin drivers, concentration, China)

DESIGN · 5 · PATTERN

Pipeline: four sequential sub-agents, each reading only the previous file(s). Small contexts, an auditable trail.

DESIGN · 6 · OUTPUTS & GATES

financials.md → trends.md → exec-summary.md → dashboard.html. Nothing is sent; I check the numbers.

DEMO 4 · WORKFLOW

Four stages, handed off through files

Annual report
10K-NVDA.pdf, about 100 pages

→

1 · Extract
financials.md: every figure with its page

→

2 · Analyse
trends.md: YoY, vs guidance, anomalies

→

3 · Summarise
exec-summary.md: one page for the Board

→

4 · Dashboard
dashboard.html: view as Board, CFO or investor

SMALL CONTEXTS
Each stage reads one file, not 100 pages plus a conversation.

AUDIT TRAIL
Every number traces to a page; calculations are marked.

RE-RUN ONE STAGE
Change the summary without re-extracting.

DEMO 4 · BEAT 5 · HARDEN

Extract → analyse → summarise → dashboard (1/2)

Process NVIDIA's FY2026 annual report (10K-NVDA.pdf) in 4 sequential sub-agents to avoid filling the context window. Each sub-agent reads only its assigned input file(s). Complete and save each stage before starting the next. Save all files to Documents/Cowork/NVDA-FY26/.

SUB-AGENT 1 — Extract financial figures Reads: 10K-NVDA.pdf · Output: financials.md
Extract all financial figures and KPIs, with page numbers. Cover revenue by reportable segment (Compute & Networking, Graphics) and by market platform (Data Center compute and networking, Gaming, Professional Visualization, Automotive, OEM & Other); gross margin and its drivers (inventory provisions, H20 charges); R&D; SG&A; operating income by segment; other income; effective tax rate; net income; diluted EPS; operating and free cash flow; buybacks and dividends; cash and marketable securities; inventories; purchase obligations and supply commitments; customer concentration; revenue by geography. FY2026 and FY2025.
Organize the figures into structured tables. Preserve each metric's reported name, period, segment, unit, page reference and restatement notes. Record both the printed page and the PDF page index when they differ.

SUB-AGENT 2 — Analyze trends and anomalies Reads: financials.md · Output: trends.md
Identify YoY trends, results vs. guidance, and anomalies — e.g., revenue up 65% while gross margin fell; net income growing faster than operating income; rising customer concentration; inventory growing faster than revenue; one-offs; restated figures. Show formulas and inputs. Separate the report's explanations from your interpretations.

SUB-AGENT 3 — Write the executive summary Reads: trends.md · Output: exec-summary.md
Write a 1-page executive summary for the Board: 5 headlines, 3 risks, 3 questions management should answer. Keep page references for all numerical claims.

DEMO 4 · BEAT 5 · HARDEN

Extract → analyse → summarise → dashboard (2/2)

SUB-AGENT 4 — Create the dashboard Reads: financials.md, trends.md, exec-summary.md · Output: dashboard.html

Create a professional, interactive dashboard in one self-contained HTML file with embedded CSS, JavaScript and data. It must open locally in a browser without installation, a server or an internet connection. Include:

Executive snapshot: KPI cards for revenue, gross margin, operating income, net income, EPS, free cash flow, cash and inventories — this year, last year, change and source page.
Financial performance: dollars and ratios on separate charts; quarterly revenue and gross-margin trend; results vs. guidance.
Segments and platforms: Compute & Networking vs. Graphics with a segment filter that updates cards, charts and tables; revenue by market platform. Flag group-only metrics.
Board insights: the 5 headlines, 3 risks and 3 questions, plus anomalies from trends.md with evidence.
Evidence explorer: searchable, sortable table of metric, period, segment, value, unit, page and reported/calc status, with formulas for calculated values.
For my role: a "View as" selector (Board / CFO / Investor) that reorders the KPI cards. CSV export and a clean print layout.

RULES

Every number carries its page reference, including numbers in cards, charts and tables. Mark anything you calculated "calc" and show the formula and inputs.
If a figure is not in the report, write "not disclosed." Never substitute zero, estimate or invent guidance.
Distinguish % growth from percentage-point changes in ratios. Don't round away movements under 0.5 points.
Flag one-offs (e.g., H20 charges), unrealised investment gains and changes in non-GAAP definitions.
Before delivery, reconcile the dashboard against financials.md and trends.md, and test the filter, CSV export and print layout.

DEMO 4 · BEAT 4 · RESULTS

NVIDIA FY2026: performance, concentration and commitments

SCALE AND MARGIN

Revenue $215.94B, up 65.47%; GAAP gross margin 71.1%, down 3.90 points. Management cites H20 and Blackwell; an exact causal split is not disclosed.

CUSTOMER CONCENTRATION

Two anonymous direct customers represent 22% and 14% of revenue. The largest FY2025 share was 12%; continuity of customer identity is unknown.

CASH AND COMMITMENTS

Company-defined FCF is $96.58B. Inventory is $21.40B and supply commitments about $95.20B. FCF excludes major equity purchases and the Groq payment.

HISTORICAL OUTLOOK

Q4 revenue of $68.13B exceeded the $65.0B midpoint by 4.81%. Q1 FY2027 revenue guidance was $78B ±2%, assuming zero China Data Center compute revenue.

Explore the dashboard: Board / CFO / Investor views · segment filters · source pages · CSV export

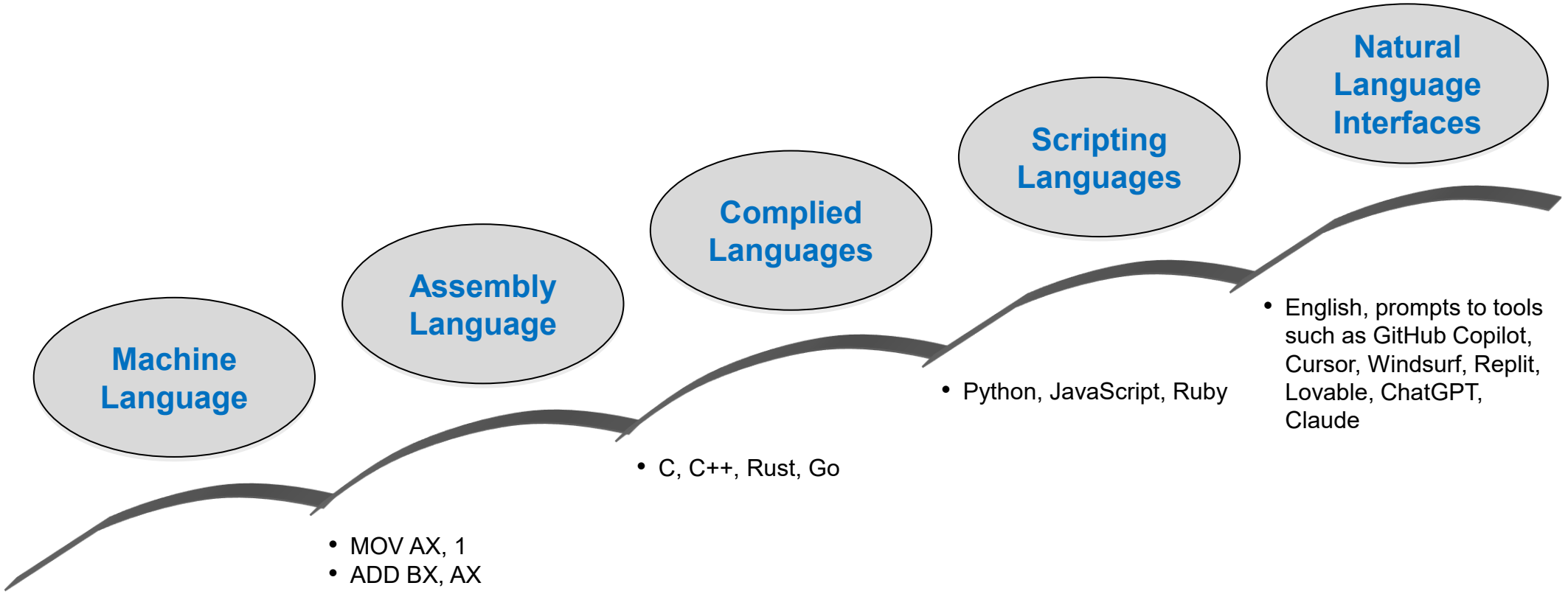
Sources: NVIDIA FY2026 10-K pp 41, 51, 69, 70, 78 and Q3/Q4 earnings releases. Year ended 25 Jan 2026; historical outlook issued 25 Feb 2026.

VIBE CODING TO AGENTIC ENGINEERING

VIBE CODING: EVERYONE CAN BE A PROGRAMMER



EVOLUTION OF CODING LANGUAGE



Examples

- 10110100, 11001010 (binary opcodes)
- MOV AX, 1
- ADD BX, AX
- C, C++, Rust, Go
- Python, JavaScript, Ruby
- English, prompts to tools such as GitHub Copilot, Cursor, Windsurf, Replit, Lovable, ChatGPT, Claude

- The earliest and most basic level of programming, consisting of **binary code executed directly by the CPU.**
- A **symbolic representation** of machine code using mnemonics.
- **High-level languages** that are **translated (compiled)** into machine code before execution
- **High-level, often interpreted** languages designed for automation, rapid development, and ease of use
- Use of **human languages** to instruct computers, often via **AI models** that interpret intent and generate or execute code. Still **emerging but increasingly powerful**.

VIBE CODING TOOLS BY LEARNING CURVE

Vibe coding tools have a different learning curve, environment, and degree of autonomy

Low **Medium** **High**

- **Prompt-based coding tools**
  - **Replit**
  - **Lovable**
  - **Base44**
  - **Vercel**
  - Google AI Studio
  - Bolt
  - GitHut Spark
- Treat Coding like **conversational creation**
- Best for prototypes, MVPs, and demos
- High abstraction
- Limited team workflows

- **IDE-based coding tools**
  - **GitHub Copilot**
  - **Cursor**
  - **Antigravity**
  - CLINE
  - Windsurf
  - Plug-in: Claude Code, Codex
- Treat coding like **augmented software engineering**
- **Ask / Plan / Agent** mode
- **Easier for developers review** the code change and accept / reject
- Strong team workflows (Git, PRs, reviews)
- Coding agent can make some decisions, choose tools
- Most of tools predefined

- **CLI-based coding agent**
  - **Claude Code**
  - **GitHub Copilot CLI**
  - **GPT-5.3-Codex**
  - Open Code
  - Gemini CLI
- **Task oriented** and **highly agentic**: treat coding like **agentic software engineering**
- Strong for **automation** : plan, critique, revise/edit
- Use **skills, sub agents, agent teams** for more structure, better reproducibility and **parallel execution**,
- **Markdown file** as memory
  - CLAUDE.md / Agents.md: eager
  - Skills.md: lazy (invoked when needed)
- Unix style POSIX-compatible shell command (Bash)
  - Open, create, modify the file
- Can create new tools on the fly (e.g., writing a test)

VIBE CODING TOOLS: SIDE-BY-SIDE COMPARISON

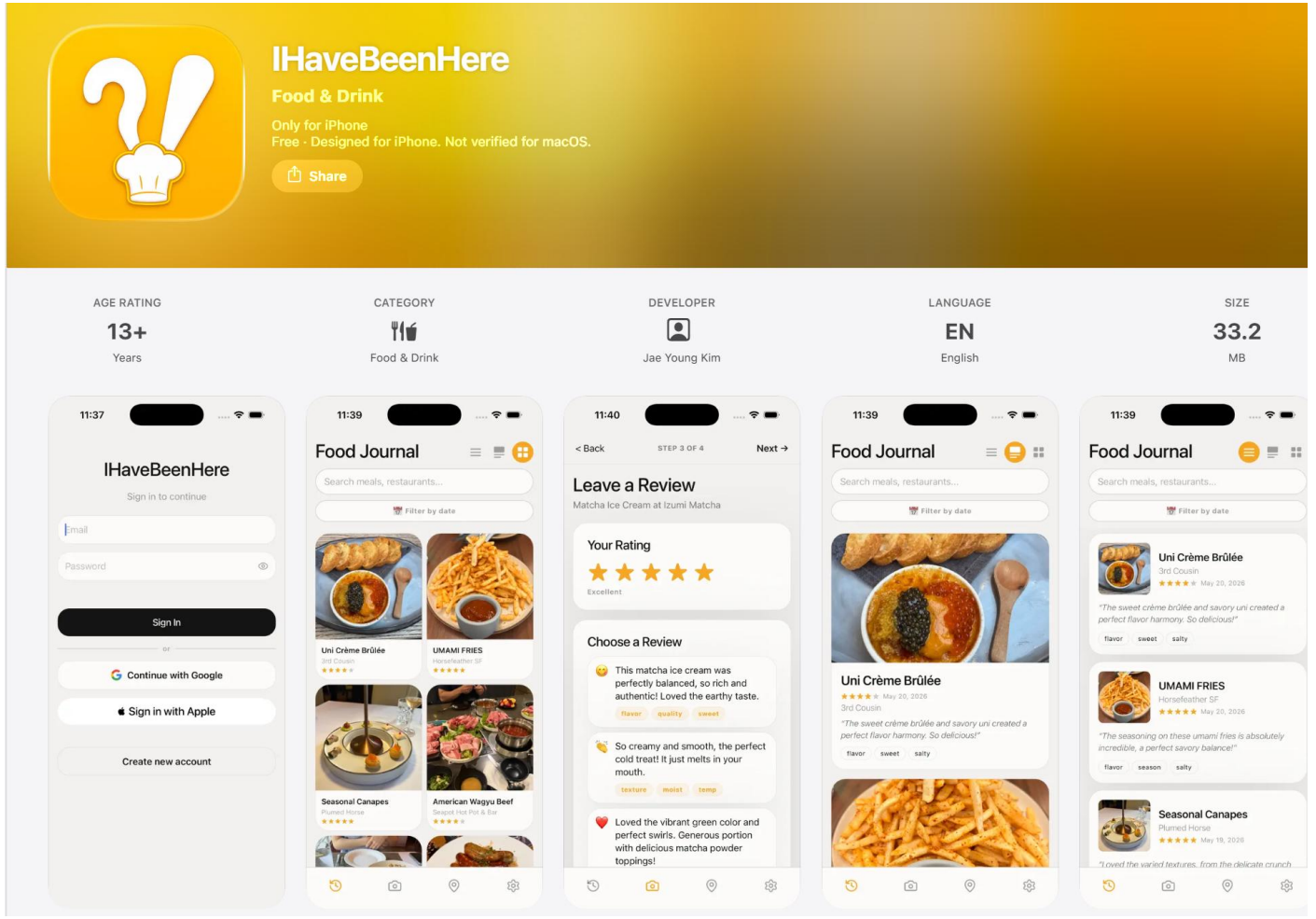
Feature	Prompt-Based Tools (Replit, Base44, Lovable)	IDE-Based Tools (Cursor, GitHub Copilot, Antigravity)	CLI-Based Coding Agent (Claude Code, GitHub Copilot CLI)
Primary interface	Chat	Code editor	Terminal (CLI)
Best for	Prototypes, MVPs	Production code, research	Production code, Automation, debugging, experiments
Control level	Low-medium	High	Developer-defined
Architecture	AI-generated	Developer-defined	Developer-defined
Refactoring	Weak	Strong	Very strong (automated)
Environment access	Limited	Editor-level / Selected shell access under "agent" mode	Full shell access
Execution	No	No / Yes under "agent" mode	Yes (runs commands)
Learning curve	Very low	Moderate	Moderate-high
Team workflows	Limited	Strong	Strong (scriptable)
Deployment	Often one-click	External	Scriptable via CLI
Who it is for	Beginners, founders	Engineers, researchers	Power users, ML/infra engineers

LIVE DEMO


CLAUDE
CURSOR
1
2
3 def greeting(name: str):
4 printc("Hello, ("World'))
5
6
7
8
9
WINDSURF

MOBILE APPS PRODUCED BY VIBE CODING:

BAY K-AI VIBE CODING PROJECT STUDY GROUP EXAMPLE



https://apps.apple.com/us/app/ihavebeenhere/id6771244099

VIBE CODING WITH GOOGLE AI STUDIO: AI BROCHURE APP

Create AI Brochure Maker based on Nano Banana Image Editing Model

(1) Users will upload product image, human model image, and background image, and type advertising copy.
(2) Create 4 separate input blocks in the UI and put nice title.
(3) Use an image editing model to combine these elements to generate a brochure image and show on the UI
(4) Let users be able to download generated brochure image file if they want

Put 6 default images for fast tests of the app: product image

AI Advertising Maker

VIBE CODING 101 – GAME APP FROM SINGLE PROMPT** (1/5)

(1) Brick Breaker
Make a brick breaker game. There's a paddle at the bottom that follows my mouse, a ball that bounces around, and rows of colorful bricks at the top that disappear when the ball hits them. Show my score and how many lives I have left. Give me 5 lives. Add a start screen and a "you win" screen. Make it look bright and arcade-y, and add a little sparkle burst when a brick breaks. Add a button that makes the paddle wider

(2) Memory Match
Make a memory matching game with 16 cards face down. I flip two at a time and if they match they stay up. Use fun emoji on the cards. Count my moves and celebrate when I clear the board.

(3) Snake
Make the classic snake game. Arrow keys to move, eat the dots to grow, don't hit the walls or yourself. Show the score and a game over screen with a restart button. Make it look like an old Nokia phone.

(4) Two-player Pong
Make a two-player pong game on one keyboard. One person uses W and S, the other uses the up and down arrows. First to 5 wins. Big retro scoreboard in the middle.

**Examples from Minha Hwang

VIBE CODING 101 - FUN APP FROM SINGLE PROMPT** (2/5)

(5) Name Spinner
Make me a spinning wheel that picks a random name. I want to type in a list of names and hit spin. Make it actually spin and slow down, with a drumroll feeling, and confetti when it lands on someone.

(6) Countdown
Make me a countdown to December 25th. Big numbers, days hours minutes seconds, ticking live. Make it feel festive and cozy. When it hits zero I want fireworks.

(7) Meeting Cost Ticker
Make a timer that shows how much money a meeting is costing while it runs. I type in how many people are in the room and their rough hourly rate, hit start, and watch the dollar amount climb.

(8) Silly Quiz
Make a 5-question quiz called "What kind of coffee are you?" Fun questions, four choices each, and at the end tell me my result with a description and an emoji. Make it colorful.

**Examples from Minha Hwang

VIBE CODING 101 - UTILITY APP FROM SINGLE PROMPT** (3/5)

(9) Bill Splitter

Make something that splits a restaurant bill. I put in the total, how many people, and how much tip, and it tells me what each person owes. Make it look nice, not like a calculator.

(10) Throw Messy Data

Build a dashboard where I paste in CSV data or upload a file, and it automatically detects the columns, guesses which are dates/numbers/categories, and generates the right charts. Let me pick which column to group by. Include summary stats at the top and a sortable table below. Handle missing values gracefully instead of crashing.

(11) Interactive Compound Interest Visualizer

Build me an interactive compound interest visualizer. Sliders for starting amount, monthly contribution, annual return %, and years. Show a line chart that updates live comparing total contributions vs. total value, plus a big headline number for the final balance and how much of it was growth. Warm, friendly design — rounded cards, one accent color. Add a preset button for "start at 25 vs. start at 35" so I can see the gap.

**Examples from Minha Hwang

VIBE CODING 101 - WEB APP FROM SINGLE PROMPT* (4/5)

(12) Fireworks

Build an interactive fireworks show. When I click anywhere on the screen, launch a firework from the bottom to that point and explode it into colorful particles with realistic gravity and fade. Add a dark night sky background with stars. Include a button to launch a random automatic show of multiple fireworks.

(13) Color Palette Generator

Build a color palette generator. Let me pick a base color with a color picker or hex input, then show 5-color harmonies (complementary, analogous, triadic, tetradic, monochromatic) I can switch between with tabs. Tapping any swatch copies its hex code to the clipboard with a toast confirmation. Include a shuffle button that picks a random pleasant base color. Make it clean and modern — the kind of thing you'd actually open on your phone.

* From Andrew Ng

VIBE CODING 101 - WEB APP FROM SINGLE PROMPT* (5/5)

(14) Pomodoro Timer

Build a 25-minute pomodoro timer with a circular progress ring that drains as time passes. Include start, pause, and reset buttons. When time hits zero, play a short beep and show a "Break time!" message.

(15) Memory Card Matching Game

Build a memory card matching game. Show a 4x4 grid of face-down cards with emoji pairs. Click to flip cards — if two match they stay face up, otherwise they flip back. Track the number of moves and time elapsed. Show a congratulations message when all pairs are found. Add a restart button.

* From Andrew Ng

FINAL THOUGHTS

FIRST-HAND, HANDS-ON EXPERIENCE

(1) Use (or Build) a simple agent for your personal use case: no-code / low-code tools (prompt only)

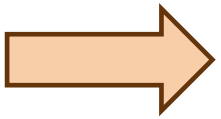
- Sub Agents or Agent Teams in Claude Code or GitHub Copilot CLI:
- Genspark - Your All-in-One AI Workspace
- Manus: Hands On AI

(2) Generate multimedia content by using generative AI tools

- Voice / Music / Sound Effect: ElevenLabs - https://elevenlabs.io/
- Image: Gemini Nano Banana Pro - Nano Banana Pro: Gemini 3 Pro Image model from Google DeepMind
- Video: Sora,Veo3 - Sora | OpenAI, Veo 3 | Google AI Studio

(3) Try vibe coding to create a simple app

- Claude.ai - Interactive Artifact for instant check and run: https://claude.ai/
- Antigravity – free vibe coding tool from Google Gemini: Google Antigravity Blog: introducing-google-antigravity
- GitHub Copilot: GitHub Copilot · Your AI pair programmer · GitHub
- Claude Code: Command line interface (CLI) coding agent: Claude Code - AI coding agent for terminal & IDE | Claude
- Google AI Studio – featuring feature suggestion and one click web app deployment: https://aistudio.google.com/
- Replit: https://learn.deeplearning.ai/courses/vibe-coding-101-with-replit
- Lovable: Lovable - Build Apps & Websites with AI, Fast | No Code App Builder



Learning by doing: Build a grounded understanding of AI's key capabilities and limitations through hands-on experience.

Don't rush to conclusions: AI is evolving rapidly—what was true before agentic coding breakthroughs (e.g., Feb 2025) may no longer hold today.

FOCUS ON LEARNING FUNDAMENTAL SKILLS

	Description	Key Quotes
Strategic Vision Setting	Strategic vision-setting remains beyond AI's capabilities, emphasizing that humans must define ambitious goals and inspire others to pursue them.	• What can the models not do? Aspire. Set the right aspiration. Do you go to low Earth orbit? Do you go to the moon? Do you go to Mars? That's a uniquely human capability.
Judgement	Humans must establish ethical boundaries and operational parameters based on values and societal norms that AI cannot inherently understand.	• There's no right and wrong in these models, and so how do you set the right parameters?
(True) Creativity	AI can predict patterns from existing data, while humans can think orthogonally and develop entirely novel approaches outside established frameworks.	• The models are inference models — the next most likely step.
Critical Thinking	AI can support your thinking, but you still own the final decision. Critical thinking is what keeps humans in control—separating *plausible* from *true*, and *helpful* from *harmful*.	• Analytical thinking remains the most sought-after core skill among employers, with seven out of 10 companies considering it as essential in 2025.
Communication	Humans are still in charge of communication. Empathy, listening, negotiation, conflict resolution, persuasion—these are "relationship bandwidth" skills that AI doesn't replace.	• Communication is the most important skill in life. • Mastering soft skills like communication and critical thinking may be even more crucial than technical know-how.

McKinsey Boss Shares 3 Human Skills That AI Models Can't Do - Business Insider
I'm an AI strategist. Here are 4 things you'll never catch me using AI for ; Human skills will matter more than ever in the age of AI | McKinsey